

Oisin McSherry

Software Engineer

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Software engineer with around three years' experience across full-stack development, applied machine learning, and data-heavy systems, spanning an industry internship, university research, and full-time roles. Currently the sole developer for an automotive parts business, where I rebuilt a 2.4-million-part pricing and inventory system end to end. First Class MSc in Computer Science. I build tools that run in production and hold up under real use, from internal pricing platforms to live, ML-driven projects with working demos.

Technical Skills

Languages	Proficient: Python, JavaScript/TypeScript, SQL, Java, Kotlin Familiar: C, C#, Swift, PHP, MATLAB, HTML/CSS, Bash
Frameworks & Tools	Android Studio, Xcode, React, Flask, Docker, Maven, Gradle, JUnit, Git, GitHub, Arduino, Blender, VS Code, IntelliJ, PyCharm
Databases & Cloud	MySQL, SQLite, MongoDB, Neo4j, AWS Elastic Beanstalk, CoreData, dBeaver
Specialisms	Full-Stack Development, Applied Machine Learning, AI Agents & LLM Integration, Data Visualisation, Mobile Development, HCI & UX, Cloud Computing, Web Scraping

Employment History

Software Developer

Aug 2025 - Jul 2026

Premier Auto Parts

Cork, Ireland

- Sole developer for the business, owning the full stack end to end, from data infrastructure and pricing systems to internal tools and dashboards.
- Cut a bloated 2.4-million-part product database (two decades of accumulated backlog, duplicates, and dead stock that were slowing the system and misrepresenting stock levels and valuations) down to a clean core of around 140,000 parts, then rebuilt it to roughly 400,000 by adding current-range parts that had been missing. Correctly valued across dozens of brands and suppliers, fixing data that was directly distorting costing, margins, and stock value.
- Built a full-stack pricing application that automatically gathers competitor prices, demand signals, new product listings, and rebate data from around a dozen approved sources, replacing a manual process that took roughly a week every two months, and feeding rebate (up to ~30%) and competitive pricing decisions.
- Wrote batch processing to apply grouping and markup rules across the full parts catalogue, and built dashboards giving management clearer visibility into sales, purchasing, and stock valuation. Across my time there, these data and tooling improvements contributed to a 30%+ increase in sales.
- Designed and built internal tools for invoice and credit processing, streamlining accounts workflows and cutting manual effort across the business.
- Handled IT responsibilities across the small company including system support and infrastructure maintenance.
- Primary technologies: Python, React, SQL (MySQL/SQLite), Flask, web scraping libraries; selected front-end and database tooling based on what was most appropriate and in-demand for each deliverable.

Research Software Developer

Jun 2025 - Present

University College Cork - Dark Design Patterns Project

Cork, Ireland

(Active Jun-Nov 2025; paused pending funding; re-engaged Apr 2026 following successful proposal progression)

- Contributed to academic research into dark design patterns in social media, with a focus on how persuasive technology affects user behaviour.
- Built Python tooling for automated data gathering, cleaning, sorting, and visualisation; managed an SQL database of research papers and collected datasets.
- Developed a native Android application (Java/Kotlin) to collect device usage data from participants, capturing metrics such as screen time and per-app usage.
- Engineered a social media recreation application with three distinct versions: Light (dark patterns removed to reduce addictive behaviour), Neutral (moderate engagement mechanics similar to modern apps), and Dark (full dark pattern implementation to maximise user retention and reduce exit ease), used as stimuli in user studies.

Teaching & Mentorship

University College Cork

Jun 2024 - Sep 2025
Cork, Ireland

ICT Summer Camp Demonstrator

- Taught and mentored students, introducing programming concepts through interactive lessons and hands-on coding exercises.
- Guided students through debugging and error resolution, reinforcing best coding practices.

Munster Programming Training (MPT) Demonstrator

- Instructing senior secondary school students in Python and web development, developing computational thinking and problem-solving skills.

UCC MScIM Teaching Assistant

- Instructing masters students in Python, Swift, and Chuck.

Software Engineer

Jun 2024 - Feb 2025
Cork, Ireland

Tyndall National Institute

- Developed an Android-based fatigue monitoring system integrating Bluetooth Low Energy (BLE) connections to collect real-time data from pressure insoles and IMUs, optimising data accuracy and power efficiency.
- Integrated a MATLAB computational model into the Android app to enable real-time motion data processing and visualisation, replacing reliance on post-experiment CSV exports.
- Engineered a C-based application for wearable sensors using cyclic voltammetry for measuring electrochemical properties in biological samples.
- Designed and implemented an interactive 3D sensor visualisation interface, reducing sensor calibration errors by 40% and significantly improving research efficiency.

Software Engineer Intern

Mar 2022 - Sep 2022
Ireland

Cadence Design Systems

- Developed dynamic SQL queries and Python scripts to extract, clean, and visualise company data, improving efficiency in backlog tracking and resource allocation.
- Designed and presented interactive data dashboards enabling management to identify bottlenecks in workload distribution and optimise task assignments.
- Developed and deployed an internal web application for company-wide self-service data access, enabling employees to retrieve key insights without requiring SQL knowledge.

Education

MSc Computer Science (Interactive Media) - First Class Honours

Nov 2023 - Nov 2024

University College Cork

BSc Computer Science - Upper Second Class Honours

Sep 2019 - Jun 2023

University College Cork

Selected Projects

DarkScan - Dark Pattern Detector

github.com/Oisin33/DarkScan

A browser-extension and ML system that detects manipulative “dark pattern” UI, building on my UCC dark-patterns research. Uses a tiered classifier cascade (heuristics → TF-IDF → embeddings → LLM) that generalises to unseen patterns and learns from user feedback. The trained model is exported to run live in the browser. Python, scikit-learn, Flask, React.

Nexus - AI Codebase Q&A

github.com/Oisin33/Nexus

An AI agent that answers natural-language questions about any GitHub codebase by reading the code, searching, opening files, and following references via a tool-calling loop over a BM25-indexed repository, with real token streaming. FastAPI, Claude API, TypeScript/React, Docker.

Dark Patterns Research App

Three-variant Android social media recreation (Light/Neutral/Dark) used as experimental stimuli for academic user studies on persuasive technology. Built in Java/Kotlin.

Tyre Pricing Platform

Full-stack web application that scrapes approved sources to build a live pricing and demand database for a tyre retail business, including rebate tracking and new product discovery.

MSc Thesis - Fatigue Monitor

Android application integrating wearable BLE sensors with real-time MATLAB-driven motion visualisation, reducing calibration errors by 40%.

VR Web Application	Interactive 3D lecture hall built with Three.js, incorporating animated models for an immersive educational experience.
Usability Engineering	Redesigned a poorly structured website applying modern UI/UX and accessibility principles.
2D Platformer Game	Browser-based game built with JavaScript and HTML5 Canvas, implementing custom collision detection and physics.

Interests

Science fiction and fantasy literature • Grand strategy and 4X gaming • Squash and tennis